

From Rate Suppression to Causal Gradient Starvation: A Finite-Width Theory of Recurrent Learning

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Abstract

A strong predictive cue can accelerate optimization while preventing a weaker but independently predictive cue from being learned. We study this gradient-starvation question in recurrent binary classifiers through a paired intervention: a both- feature trajectory is compared with a same-initialization trajectory in which only the strong input channel is ablated. The contribution is a finite-width package, not a claimed general dynamical mean-field theory (DMFT). We derive exact signed- logit and response dynamics through the full sample empirical NTK and response- logit cross-kernel. Freezing these objects defines a nonlinear logistic tangent surrogate; continuous Grönwall and experiment-matched discrete secant-kernel bounds compare it with nonlinear training only under supplied kernel-movement control. For arbitrary sign changes, cumulative negative relative drift exceeds cumulative positive drift exactly when strict outcome suppression occurs. Under a single transfer-to-rate-suppression crossing this becomes the tail-area criterion, but causal starvation additionally requires an independent, preregistered weak-only first-hit learnability gate. A noiseless rank-one log drift ratio gives sufficient conditions for one rate crossing. In the dense-linear model, the realized initialization also gives an exact pre-trajectory rate-gap threshold; uniform product-ball bounds, when independently proved, imply crossing, outcome, learnability, or finite-horizon safety, but no such nonvacuous positive-disorder constants are currently constructed. We also prove stability of quantitatively isolated transverse first hits under uniform-error, radius, and mesh hypotheses, an exact six-scalar zero-disorder/zero-lag wide limit whose symmetric reference is suppressed from initialization while its weak-only trajectory remains learnable, and—when feasible—the fixed-chart, uncapped, target-attaining minimum-norm property and local one-step bound of Counterfactual Drift Correction (CDC).

Controlled dense-linear experiments show positive weak-learning AUC gaps in all 32 ρ -seed cells per lag at lags 0, 2, and 4 (four strengths by eight seed IDs). In the final eight-seed-per-architecture nonlinear study, tanh has 3/8 causal certificates; GRU has 5/8 response crossings but 0/8 causal certificates because the weak-only model never reaches $\beta = 0.5$. A post-hoc terminal sensitivity audit, derived without retraining, places $\tanh B_W(H)$ in [4.413, 4.676] and GRU in [0.0828, 0.187]; it is not a replacement for the first-hit gate or an architecture ranking. A sealed five-endpoint frozen-kernel pilot fails its preregistered rule. Only after that failure, a fresh restricted 4×8 data/model-seed factorial classifies response-crossing events in 64/64 records and drift-crossing events in 60/64, with two-way-bootstrap 95% lower endpoints 1.0 and 0.8125. These are one-cell event-classification results, not phase, causal, learnability, calibrated-time, magnitude, prevalence, or kernel-stability validation. All executed empirical evidence is synthetic and seed-limited. A semi-real generated-cue intervention and expanded nonlinear/CDC tradeoff study are frozen but unauthorized and outcome-free; neither contributes an empirical claim. A general positive-disorder/positive-lag optimization-time DMFT remains open.

1 Introduction

Cross-entropy training can favor a subset of predictive features strongly enough that other useful features are learned slowly or not at all. This phenomenon was formalized as gradient starvation by Pezeshki et al. [2021]. Recurrent networks add a temporal geometry: cue strength, lag, shared recurrent parameters, and evolving feature response geometry all affect which signal receives gradient. The resulting mechanism is easy to overstate. A weak-feature drift can decline without changing sign; its relative drift can change sign without the weak response falling behind; and a response gap is not evidence of causal starvation if the weak-only counterfactual is itself unable to learn.

We address these distinctions with a paired intervention. The both-feature and weak-only models share initialization, labels, samples, weak coordinates, and noise; only the strong input channel is set to zero in the weak-only condition. We track common feature-response functionals in both trajectories. This setup separates four statements:

1. *relative transfer*: the strong cue initially increases the weak response rate, so $d(\tau) > 0$;
2. *rate suppression*: the both-feature weak response later grows more slowly than its counterfactual, so $d(\tau) < 0$;
3. *outcome suppression*: the both-feature weak response falls below its matched counterfactual, so $\Delta(\tau) < 0$; and
4. *causal starvation*: outcome suppression occurs while the weak-only counterfactual demonstrably learns under the preregistered first-hit gate.

A transfer-to-starvation trajectory additionally has an initial positive excursion before outcome suppression. The distinction is empirical as well as logical: the final GRU study has response crossings but fails the causal learnability gate in every seed.

Contributions.

1. We prove exact finite-width logistic response flow and full signed-logit empirical-NTK/response dynamics. A closed projected equation $\dot{M} = Gg$ requires exact parameter-independent logit realization; the initialization-frozen nonlinear logistic surrogate is exact internally, while comparison with a trained nonlinear model is conditional on supplied kernel-movement bounds.
2. We prove a cumulative drift-balance proposition for arbitrary sign changes, with the exact single-crossing tail-area criterion as a corollary. We add a conditional noiseless rank-one rate-crossing criterion and an exact realized-initialization dense-linear rate-gap threshold. Independently proved product-ball bounds then imply unique crossing, outcome suppression, optional weak-only learnability, or finite-horizon safe transfer; no current component constructs these bounds for positive-disorder RNNs. We also prove stability of quantitatively isolated transverse first hits under uniform-error, radius, and mesh hypotheses, and an exact zero-disorder/zero-lag six-scalar wide limit with immediate deterministic suppression and weak-only learnability.
3. We operationalize the rate–outcome–causal hierarchy through paired first-hit, AUC, direct-drift, and finite-step certificates. A post-hoc terminal β profile reports threshold sensitivity from existing outcomes without replacing the preregistered first-hit gate.
4. We report the full empirical sequence rather than only positive endpoints: corrected dense E1, seed-dependent nonlinear E-NL, a failed five-endpoint E19 pilot, and the fresh E20 success restricted to drift/response event labels. The archives separate auditable outcomes from historically unreconstructible source.
5. In the fixed implemented Euclidean/Frobenius chart, we prove CDC’s exact instantaneous strong-drift preservation, feasible uncapped target-attaining minimum-norm property, and local $O(\eta^2)$ one-step bound. The ablation shows this local distinction without establishing broad practical superiority or best final retention.

Claim boundary. We do not claim that recurrent networks universally undergo transfer-to-starvation, that a general joint CE-RNN optimization-time DMFT has been solved, or that an analytic ρ_c or universal (ρ, lag) phase boundary is known. The exact initialization threshold is realization-dependent at positive lag, and its trajectory implications require uniform product-ball constants that are not currently constructed. The frozen empirical-NTK surrogate is exact for its initialization-linearized tangent model, not for trained tanh/GRU networks: no required instantaneous or secant-kernel stability bound is proved here. E19 is a failed broad pilot. E20 validates only binary drift/response-crossing classification at one width/task cell; it does not validate phase, causal starvation, weak-only learnability, calibrated trajectories, crossing times, magnitudes, or prevalence. The eight-seed nonlinear proportions are descriptive, not population prevalence. All executed experiments are controlled and synthetic. The frozen semi-real and expanded nonlinear/CDC

protocols remain unauthorized and outcome-free, so they add no empirical claim. The general DMFT solver remains deliberately unimplemented where proof obligations are unmet.

2 Related work

Gradient starvation. Pezeshki et al. [2021] identify feature starvation under cross-entropy and motivate Spectral Decoupling. Our focus is different: recurrent temporal features, a paired strong-channel intervention, exact finite-width response dynamics, and the logical separation among rate suppression, outcome suppression, and causally gated starvation.

Shortcut learning and spurious-correlation robustness. Shortcut learning and simplicity bias explain why predictive but unintended cues can dominate learned representations [Geirhos et al., 2020, Shah et al., 2020]. Group DRO, Just Train Twice, and last-layer retraining instead improve worst-group performance under spurious correlations [Sagawa et al., 2020, Liu et al., 2021, Kirichenko et al., 2023]. These phenomena motivate our setting, but they do not supply the paired response-level causal criterion used here. We are not proposing a competing robustness algorithm: our diagnostic requires a stronger matched-shadow information setting, and all current evidence is synthetic.

Learning dynamics in recurrent and wide networks. Neural tangent kernels describe infinite-width gradient dynamics [Jacot et al., 2018], and wide-network linearization motivates frozen-tangent comparisons [Lee et al., 2019]. Our initialization-frozen logistic surrogate retains the nonlinear loss but is exact only for its tangent model. Its comparison with a trained finite-width network is conditional on explicit instantaneous or secant-kernel movement bounds, which we do not prove for tanh or GRU. Ger and Barak [2026] derive low-dimensional overlap learning dynamics for low-rank RNNs and show that overlaps invisible to the loss can retain training history. This directly cautions against closing a recurrent learning theory using only visible response coordinates. Clark et al. [2026] develop a mean-field account of structure and disorder in task-trained recurrent circuits in a long-time Langevin/Gibbs setting. That equilibrium construction is distinct from the paired, noiseless optimization-time gradient flow studied here. Accordingly, we claim neither the first learning theory for RNNs nor a completed general RNN DMFT; our novelty lies in the causal pairing and finite-width rate-to-outcome results.¹

Gradient surgery. PCGrad projects conflicting task gradients [Yu et al., 2020]; Bloop stabilizes an auxiliary projection using an exponential moving average [Hsieh et al., 2024]. Projection itself is not our novelty. CDC protects the gradient of a theory-defined strong-feature response and takes its target from a matched causal weak-only shadow. This stronger information setting must be reported: CDC requires a constructible ablation, a second model, lockstep training, and full-batch gradients.

3 Causal setup and definitions

3.1 Paired logistic gradient flow

For condition $C \in \{\text{B}, \text{W}\}$, let $\theta_C(\tau) \in \mathbb{R}^P$ follow full-batch gradient flow on signed logits $r_i^C(\theta) = y_i f_C(x_i; \theta)$:

$$L_C(\theta) = \frac{1}{n_C} \sum_{i=1}^{n_C} \log(1 + e^{-r_i^C(\theta)}), \quad \dot{\theta}_C = -\nabla L_C(\theta_C). \quad (1)$$

The two conditions share initialization and all data except that the strong channel is zeroed under W. Let M_s^C, M_w^C be common differentiable strong and weak response functionals. Shared initialization gives

$$\Delta(0) = 0, \quad \Delta(\tau) = M_w^{\text{B}}(\tau) - M_w^{\text{W}}(\tau). \quad (2)$$

¹The descriptions of the 2026 sources were paraphrased from the linked public sources. Content was rephrased for compliance with licensing restrictions.

Optimization time τ is distinct from sequence index k .

The controlled dense linear RNN is

$$h_{k+1} = Wh_k + Bx_k, \quad f(x) = c^\top h_T, \quad (3)$$

with $W_{ij}(0) \sim \mathcal{N}(0, g^2/N)$, $B_{ia}(0) \sim \mathcal{N}(0, 1/N)$, and $c_i(0) \sim \mathcal{N}(0, 1/N)$. The width scaling of B is essential for a non-divergent projected metric. We additionally study tanh RNNs and GRUs. For nonlinear models the two responses are common odd symmetric unit-probe contrasts, $[f(x_a^+) - f(x_a^-)]/2$, not assumed additive coordinates of every data logit.

3.2 Causal labels and metrics

Let

$$d(\tau) = \Delta'(\tau) = F_w^B(\tau) - F_w^W(\tau). \quad (4)$$

We call $d > 0$ relative transfer and $d < 0$ relative rate suppression. We call $\Delta < 0$ *outcome suppression*, whether it follows a positive excursion or is present from the start of optimization. A response crossing is the special case in which Δ returns to zero after a positive excursion and then becomes negative. We reserve *causal starvation* for outcome suppression when the weak-only trajectory also reaches a preregistered target β within horizon H . If the weak-only model does not reach β , an outcome crossing is descriptive but not a causal-starvation certificate.

We report first-hitting-time delay when both hits are observed. Formally, $\tau_W(\beta; H) = \inf\{\tau \in [0, H] : M_w^W(\tau) \geq \beta\}$, with $\inf \emptyset = \infty$; the learnability gate requires $\tau_W(\beta; H) \in (0, H]$, excluding targets met at initialization. The horizon-wide weak AUC gap is

$$\mathcal{A}_w = \int_0^H (M_w^W(\tau) - M_w^B(\tau)) \, d\tau, \quad (5)$$

computed by the logged-grid trapezoid rule. To expose rather than hide sensitivity to the single preregistered β , Section 10 also reports the threshold-free terminal response $B_W(H) = M_w^W(H)$ across seeds and its post-hoc threshold profile. Terminal thresholding is not generally equivalent to the preregistered first-hit event on a nonmonotone trajectory, so the profile is a sensitivity diagnostic rather than a replacement gate. GSI-5 remains a margin- weight diagnostic, not a definition of starvation.

4 Exact finite-width response dynamics

Proposition 1 (Universal response-logit cross-kernel flow). *At every finite width and every optimization time at which M_a^C is differentiable,*

$$\dot{M}_a^C = \frac{1}{n_C} \sum_i K_{ai}^C \sigma(-r_i^C), \quad K_{ai}^C = \langle \nabla M_a^C, \nabla r_i^C \rangle. \quad (6)$$

No width limit or two-mode realization is required. We label this elementary chain-rule identity as a proposition: its importance here is the exact finite-width scope and the causal separation it enables, not proof difficulty.

Corollary 2 (Closed projected flow under exact realization). *If the signed logits admit the exact parameter-independent representation $r_i^C = \sum_b z_{ib}^C M_b^C$, then*

$$\dot{M}_a^C = \sum_b G_{ab}^C g_b^C, \quad G_{ab}^C = \langle \nabla M_a^C, \nabla M_b^C \rangle, \quad g_b^C = \frac{1}{n_C} \sum_i z_{ib}^C \sigma(-z_i^C \cdot M^C). \quad (7)$$

If $r_i^C = z_i^C \cdot M^C + r_i^\perp$, the exact equation instead contains

$$R_a^C = \frac{1}{n_C} \sum_i \sigma(-r_i^C) \langle \nabla M_a^C, \nabla r_i^\perp \rangle, \quad \dot{M}_a^C = (G^C g^C)_a + R_a^C. \quad (8)$$

Corollary 3 (Exact full empirical-NTK dynamics). *Let $J^C(\tau) \in \mathbb{R}^{n_C \times P}$ have row $\nabla r_i^C(\theta_C(\tau))^\top$, and define*

$$\Theta^C(\tau) = J^C(\tau)J^C(\tau)^\top, \quad c_a^C(\tau) = J^C(\tau)\nabla M_a^C(\theta_C(\tau)). \quad (9)$$

Writing $q(r) = \sigma(-r)$ componentwise, the exact finite-width flow is

$$\dot{r}^C = \frac{1}{n_C}\Theta^C(\tau)q(r^C), \quad \dot{M}_a^C = \frac{1}{n_C}c_a^C(\tau)^\top q(r^C). \quad (10)$$

No width limit or response-coordinate realization is required.

Definition 4 (Initialization-frozen nonlinear logistic surrogate). *For one condition, let $\Theta_0 = \Theta(0)$ and $c_0 = c_a(0)$. Define*

$$\dot{\bar{r}} = \frac{1}{n}\Theta_0 q(\bar{r}), \quad \dot{\bar{M}} = \frac{1}{n}c_0^\top q(\bar{r}), \quad (\bar{r}(0), \bar{M}(0)) = (r(0), M(0)). \quad (11)$$

The gate remains nonlinear. This system is exact for the initialization-linearized tangent model, not for the trained nonlinear network.

Theorem 5 (Conditional continuous and secant-SGD comparison). *Use Euclidean and induced operator norms. Suppose on $[0, t]$ that $\|\Theta(u) - \Theta_0\| \leq B_\Theta$, $\|c(u) - c_0\| \leq B_c$, and both logistic weight vectors have norm at most V . Set $a = \|\Theta_0\|/(4n)$,*

$$\phi_a(t) = \frac{e^{at} - 1}{a}, \quad \psi_a(t) = \frac{e^{at} - 1 - at}{a^2}, \quad (12)$$

with $\phi_0(t) = t$ and $\psi_0(t) = t^2/2$. Then

$$\|r(t) - \bar{r}(t)\| \leq \frac{B_\Theta V}{n}\phi_a(t), \quad (13)$$

$$|M(t) - \bar{M}(t)| \leq \frac{B_c V}{n}t + \frac{\|c_0\| B_\Theta V}{4n^2}\psi_a(t). \quad (14)$$

For explicit full-batch SGD, let $\delta\theta_k = \theta_{k+1} - \theta_k$ and define exact segment kernels

$$\tilde{J}_k = \int_0^1 J(\theta_k + s\delta\theta_k) ds, \quad \tilde{\Theta}_k = \tilde{J}_k J(\theta_k)^\top, \quad (15)$$

$$\tilde{m}_k = \int_0^1 \nabla M(\theta_k + s\delta\theta_k) ds, \quad \tilde{c}_k = J(\theta_k)\tilde{m}_k. \quad (16)$$

The true increments are exactly $r_{k+1} - r_k = \eta \tilde{\Theta}_k q(r_k)/n$ and $M_{k+1} - M_k = \eta \tilde{c}_k^\top q(r_k)/n$. If $\eta > 0$ and, for every $0 \leq j < k$, $\|\tilde{\Theta}_j - \Theta_0\| \leq D_\Theta$, $\|\tilde{c}_j - c_0\| \leq D_c$, and $\|q(r_j)\|, \|q(\bar{r}_j)\| \leq V$, then the explicit-Euler version of Eq. (11) obeys

$$\|r_k - \bar{r}_k\| \leq \frac{D_\Theta V}{n}\Phi_k, \quad (17)$$

$$|M_k - \bar{M}_k| \leq \frac{k\eta D_c V}{n} + \frac{\eta \|c_0\| D_\Theta V}{4n^2}\Psi_k, \quad (18)$$

where

$$\Phi_k = \frac{(1 + \eta a)^k - 1}{a}, \quad \Psi_k = \frac{(1 + \eta a)^k - 1}{\eta a^2} - \frac{k}{a}, \quad (19)$$

with limits $\Phi_k = k\eta$ and $\Psi_k = \eta k(k-1)/2$ at $a = 0$. Apply the bounds separately to B, W and add them to bound the paired response gap. Frozen signs or brackets transfer only when their strict margins exceed the corresponding summed errors; first hitting times additionally require Corollary 14's prehistory, transversality, radius, and mesh hypotheses.

Theorem 5 is conditional: this paper does not prove the required instantaneous or secant-kernel movement bounds for tanh or GRU. A held-out event-classification result cannot substitute for that premise. In particular, the theorem is finite-width and is not a DMFT closure.

For channel-localized modes of the dense linear RNN, differentiation across the disjoint (c, B, W) blocks gives the exact finite-width geometry

$$G_{ab} = p_a^\top p_b + \mathbf{1}\{\text{ch}(a) = \text{ch}(b)\} q_a^\top q_b + \langle S_a, S_b \rangle_F, \quad (20)$$

where $p_a = W^{\ell_a} b_a$, $q_a = (W^{\ell_a})^\top c$, and

$$S_a = \sum_{j=0}^{\ell_a-1} (W^j)^\top c (W^{\ell_a-1-j} b_a)^\top. \quad (21)$$

Theorem 6 (Exact realized-initialization rate-gap criterion). *In the noiseless positive rank-one dense-linear pair, let the strong cue have strength $\rho > 0$ at lag zero, the unit weak cue have lag ℓ , and define*

$$K = G_{ws}(0) = b_s^\top W^\ell b_w, \quad (22)$$

$$A = G_{ww}(0) = \|W^\ell b_w\|^2 + \|(W^\ell)^\top c\|^2 + \|S_w\|_F^2, \quad (23)$$

$$S_w = \sum_{j=0}^{\ell-1} (W^j)^\top c (W^{\ell-1-j} b_w)^\top, \quad (24)$$

$$s_B = \sigma[-(\rho M_s(0) + M_w(0))], \quad s_W = \sigma[-M_w(0)]. \quad (25)$$

(The recurrent sum is zero for $\ell = 0$.) The exact initial equal-time weak-rate gap is

$$d_0 = s_B(\rho K + A) - s_W A. \quad (26)$$

Consequently, initial transfer, suppression, and the boundary occur respectively when

$$K \geq \frac{A}{\rho} \left(\frac{s_W}{s_B} - 1 \right) \quad \text{or equality holds.} \quad (27)$$

This is an exact pre-trajectory criterion from the realized initialization. It is not a distribution-only positive-lag phase boundary: under the stated iid scaling, K and $M_s(0)$ contribute a seed-dependent leading sign.

The proofs are in Appendix A. Equation (7) is exact for the zero-background-noise linear synthetic model. For tanh/GRU probes, we use Proposition 1's direct autograd drift as primary and report R ; projected and matched-state decompositions are diagnostics only.

5 From rate suppression to outcome suppression and causal starvation

Proposition 7 (Cumulative drift balance). *Let Δ be absolutely continuous on $[0, H]$, with $\Delta(0) = 0$ and $d = \Delta'$ almost everywhere. Write*

$$d_+(t) = [d(t)]_+, \quad d_-(t) = [-d(t)]_+, \quad P(t) = \int_0^t d_+(u) du, \quad N(t) = \int_0^t d_-(u) du. \quad (28)$$

Then $\Delta(t) = P(t) - N(t)$ for every $t \in [0, H]$. Thus strict outcome suppression occurs by H if and only if $N(t) > P(t)$ for some $t \in (0, H]$; this statement permits arbitrary sign changes and suppression from initialization. This is deliberately a bookkeeping proposition: its contribution is the causal separation it enforces, not proof difficulty.

Corollary 8 (Single-crossing tail-area criterion). *Suppose additionally that there is $\tau_d \in (0, H)$ such that $d > 0$ almost everywhere on $(0, \tau_d)$ and $d < 0$ almost everywhere on (τ_d, H) . Define*

$$A_+ = P(\tau_d) = \Delta(\tau_d) > 0. \quad (29)$$

Then $P(t) = A_+$ and $\Delta(t) = A_+ - N(t)$ for $t \geq \tau_d$. Hence a later equality is reached by H if and only if $N(H) \geq A_+$, and strict outcome suppression occurs by H if and only if $N(H) > A_+$. If $d < 0$ on each nontrivial post-crossing interval, the later equality is unique whenever it exists.

$A \rightarrow -$ drift crossing therefore locates a local maximum of Δ ; it does not force Δ to return to zero. The post-crossing negative area must repay the positive area already accumulated. Causal *starvation* further requires the independent weak-only learnability gate.

Corollary 9 (Response-crossing bounds). *Under Corollary 8, if $d(t) \leq -\kappa < 0$ on $[t_0, H]$ with $t_0 > \tau_d$, then, when the right side lies within the horizon,*

$$\tau_{\text{response}} \leq t_0 + \frac{[A_+ - N(t_0)]_+}{\kappa} \leq t_0 + \frac{A_+}{\kappa}. \quad (30)$$

If $d(\tau_d) = 0$ and $d'(t) \leq -\lambda < 0$ on $[\tau_d, \tau_d + r]$, then $N(t) \geq \lambda(t - \tau_d)^2/2$; when $A_+ \leq \lambda r^2/2$,

$$\tau_{\text{response}} \leq \tau_d + \sqrt{2A_+/\lambda}. \quad (31)$$

Theorem 10 (Conditional parameter-tube certificate). *Let $X = (\theta_B, \theta_W)$, let V be the joint paired gradient-flow field, and define $d = L_V \Delta$ and $j = L_V d$. On the radius- R product-norm ball U_R about the shared initialization, suppose uniformly that*

$$\|V(X)\| \leq S_R, \quad -\Lambda \leq j(X) \leq -\lambda < 0, \quad 0 < \lambda \leq \Lambda, \quad (32)$$

with $S_R > 0$, and set $h_R = R/S_R$. If $d_0 > 0$ and $h_R > d_0/\lambda$, there is exactly one transfer-to-rate-suppression crossing before ball exit and

$$\frac{d_0}{\Lambda} \leq \tau_d \leq \frac{d_0}{\lambda}. \quad (33)$$

If $h_R > 2d_0/\lambda$, then $\Delta(t) < 0$ for every $2d_0/\lambda < t \leq h_R$. If also the weak-only drift obeys $\dot{M}_w^W \geq \nu > 0$ uniformly and

$$\frac{[\beta - M_w^W(0)]_+}{\nu} < h_R, \quad (34)$$

then the weak-only member reaches β within the same certified horizon, so the outcome suppression is causally gated starvation rather than suppression alone.

Corollary 11 (Certified finite-horizon safe transfer). *On the same ball, if instead $|j(X)| \leq J$ and $d_0 - Jh_R > 0$, then for $0 < t \leq h_R$,*

$$d(t) \geq d_0 - Jt > 0, \quad \Delta(t) \geq d_0 t - \frac{Jt^2}{2} > 0. \quad (35)$$

Thus suppression is excluded on this certified finite horizon, not for all future optimization time. The executable checker fails closed when optional premises are missing; it does not estimate $S_R, \lambda, \Lambda, \nu$, or J from sampled trajectories. No current component constructs nonvacuous general positive-disorder constants.

5.1 An ordering-invariant rank-one criterion

Assume deterministic positive cues $z^B = (\rho, 1)$, $z^W = (0, 1)$, $\rho > 0$, and exact two-mode realization. Define

$$u_B = \rho M_s^B + M_w^B, \quad s_B = \sigma(-u_B), \quad (36)$$

$$u_W = M_w^W, \quad s_W = \sigma(-u_W), \quad (37)$$

$$H_B = \rho G_{ws}^B + G_{ww}^B, \quad H_W = G_{ww}^W, \quad (38)$$

$$Q_B = \rho^2 G_{ss}^B + 2\rho G_{sw}^B + G_{ww}^B \geq 0. \quad (39)$$

Theorem 12 (Log drift-ratio identity). *The weak drifts factor as $F_w^B = s_B H_B$ and $F_w^W = s_W H_W$. Where $H_B, H_W > 0$, define*

$$\Psi = \log \frac{F_w^B}{F_w^W} = \log \frac{s_B}{s_W} + \log \frac{H_B}{H_W}. \quad (40)$$

Then $\text{sign}(d) = \text{sign}(\Psi)$ and

$$\Psi' = -s_B(1 - s_B)Q_B + s_W(1 - s_W)H_W + \left(\log \frac{H_B}{H_W} \right)'. \quad (41)$$

Corollary 13 (Unique rate crossing). *Suppose on $[t_0, H]$ that $H_B, H_W > 0$, $\Psi(t_0) > 0$, $\Psi'(t) \leq -\kappa < 0$, and $H - t_0 \geq \Psi(t_0)/\kappa$. Then exactly one rate crossing occurs in $(t_0, H]$ and*

$$\tau_d \leq t_0 + \Psi(t_0)/\kappa. \quad (42)$$

A sufficient primitive condition is

$$s_B(1 - s_B)Q_B \geq a, \quad s_W(1 - s_W)H_W \leq b, \quad (\log(H_B/H_W))' \leq c, \quad a - b - c \geq \kappa > 0. \quad (43)$$

The bounds must be established independently to be predictive. Evaluating Ψ on a completed trajectory is certification, not prediction. Noisy cues generally break the rank-one factorization, and even a rate crossing still requires Corollary 8 for an outcome crossing.

5.2 Hitting-time stability

Corollary 14 (Transverse first-hit stability). *Let continuous $x_N, x : [0, H] \rightarrow \mathbb{R}$ satisfy $\|x_N - x\|_\infty \leq \epsilon$, and define*

$$\tau = \inf\{t \in [0, H] : x(t) \geq \beta\}, \quad \tau_N = \inf\{t \in [0, H] : x_N(t) \geq \beta\}, \quad \inf \emptyset = \infty. \quad (44)$$

Assume x is absolutely continuous, $x(0) < \beta$, and $\tau \in (0, H)$. Choose $0 < r \leq \min\{\tau, H - \tau\}$ such that $x'(t) \geq \kappa > 0$ almost everywhere on $[\tau - r, \tau + r]$, and let

$$\eta = \beta - \max_{0 \leq t \leq \tau - r} x(t) > 0. \quad (45)$$

If $\epsilon < \eta$ and $\epsilon/\kappa \leq r$, then $\tau_N < \infty$ and

$$|\tau_N - \tau| \leq \epsilon/\kappa. \quad (46)$$

For an ordered grid $0 = t_0 < \dots < t_m = H$ of maximum mesh $h = \max_j(t_j - t_{j-1})$, let τ_N^{samp} be the first sampled time with $x_N(t_j) \geq \beta$, and let τ_N^{int} be the first hit of the piecewise-linear interpolant through the samples. If $\epsilon/\kappa + h \leq r$, both are finite and

$$|\tau_N^{\text{samp}} - \tau| \leq \epsilon/\kappa + h, \quad |\tau_N^{\text{int}} - \tau| \leq \epsilon/\kappa + h. \quad (47)$$

For a paired delay, if both reference hits are finite and each trajectory satisfies its own assumptions, then

$$|(\tau_{N,B} - \tau_{N,W}) - (\tau_B - \tau_W)| \leq \epsilon_B/\kappa_B + \epsilon_W/\kappa_W + h_B + h_W. \quad (48)$$

If either hit is infinite, the delay is censored; a common mesh h gives the special case with $2h$.

Descending crossings follow by sign reversal and localization after a strict excursion. Since the response gap has $\Delta(0) = 0$, its later downward outcome crossover must be localized to $[a, H]$ with $\Delta(a) > 0$ and treated using $x = -\Delta$ and target zero; the hit at time zero is not that later crossover. This corollary transfers an independently proved uniform trajectory limit. It does not prove that limit, first-entry semantics, the prehistory margin, or the derivative bound. Thus Theorem 5's summed paired response bound licenses a crossing-time statement only after all of these separate guards hold; an error formula or empirical event match alone is not a hitting-time certificate.

6 A solvable zero-disorder wide limit

Set recurrent gain $g = 0$, lag separation zero, and background noise zero in the dense linear model. Write the input columns as b_s, b_w . The recurrent block is inert, $M_a = c \cdot b_a$. Let $u_{ab} = b_a \cdot b_b$ and $w = c \cdot c$.

Theorem 15 (Every-width six-scalar closure). *At every width,*

$$\dot{M}_a = g_a w + \sum_b g_b u_{ab}, \quad \dot{u}_{ab} = g_a M_b + g_b M_a, \quad (49)$$

$$\dot{w} = 2 \sum_b g_b M_b, \quad G_{ab} = u_{ab} + \delta_{ab} w. \quad (50)$$

Theorem 16 (Finite-horizon deterministic wide limit). *Let all coordinates of b_s, b_w, c be mutually independent $\mathcal{N}(0, 1/N)$ and*

$$q_N = (M_s, M_w, u_{ss}, u_{sw}, u_{ww}, w), \quad q_0 = (0, 0, 1, 0, 1, 1). \quad (51)$$

For every $\epsilon > 0$,

$$\mathbb{P}(\|q_N(0) - q_0\|_\infty > \epsilon) \leq \min \left\{ 1, \frac{9}{N\epsilon^2} \right\}. \quad (52)$$

Fix a cue law with bounded second moments and define

$$g_a(M) = \mathbb{E}[z_a \sigma(-z \cdot M)], \quad A = \max_a \mathbb{E}|z_a|, \quad C = \frac{1}{4} \max_a \sum_b \mathbb{E}|z_a z_b|. \quad (53)$$

Then $\|g(M)\|_\infty \leq A$ and

$$\left\| g(M) - g(\widetilde{M}) \right\|_\infty \leq C \left\| M - \widetilde{M} \right\|_\infty. \quad (54)$$

Let $q(t)$ be the deterministic solution from q_0 , let $c_\star = W(e^{-1}) = \sup_x x \sigma(-x)$, where W is Lambert's function, and set

$$R_H = 3 + 4c_\star H, \quad B_{H,r} = R_H + r, \quad L_{H,r} = 4(A + B_{H,r}C) \quad (55)$$

for a fixed deterministic radius $r > 0$. The reference Gram energy $R = w + u_{ss} + u_{ww}$ obeys $R(0) = 3$, $R' \leq 4c_\star$, and therefore $\|q(t)\|_\infty \leq R_H$ on $[0, H]$. The vector field is $L_{H,r}$ -Lipschitz in infinity norm on the radius- r tube about this path. If

$$\|q_N(0) - q_0\|_\infty \leq e^{-L_{H,r}H} \min\{r, \delta\}, \quad (56)$$

then q_N does not leave the tube before H and

$$\sup_{t \leq H} \|q_N(t) - q(t)\|_\infty \leq \delta. \quad (57)$$

Consequently,

$$\mathbb{P}\left(\sup_{t \leq H} \|q_N(t) - q(t)\|_\infty > \delta\right) \leq \min \left\{ 1, \frac{9e^{2L_{H,r}H}}{N \min\{r, \delta\}^2} \right\}. \quad (58)$$

For $r = 1$ and $0 < \delta \leq 1$, the denominator reduces to $N\delta^2$ with the deterministic constant $L_{H,1}$. Thus $q_N \rightarrow q$ uniformly on fixed horizons in probability, with initialization scale $O_{\mathbb{P}}(N^{-1/2})$. For the paired both/weak-only result, use a common r and the maximum of the two deterministic $L_{H,r}$ constants for the two fixed cue laws.

This theorem is intentionally singular: it is a genuine large-width result only at zero disorder and zero lag. It is not a positive-disorder DMFT.

Corollary 17 (Zero-disorder suppression and weak-only learnability). *At zero bulk gain, zero lag, and zero background noise, use the deterministic symmetric initial state $(M_s, M_w, u_{ss}, u_{sw}, u_{ww}, w) = (0, 0, 1, 0, 1, 1)$. For every $\rho > 0$, the both-minus-weak gap and rate satisfy*

$$\bar{\Delta}(0) = 0, \quad \bar{d}(0) = 0, \quad \bar{d}'(0) = -\rho^2/2 < 0. \quad (59)$$

Hence $\bar{\Delta}(t) < 0$ for all sufficiently small $t > 0$: suppression begins at initialization and is not transfer-then-suppression. In the weak-only subsystem,

$$u_{ww} = w = \sqrt{1 + M_w^2}, \quad \dot{M}_w = 2\sigma(-M_w)\sqrt{1 + M_w^2} > 0, \quad (60)$$

so every finite $\beta > 0$ is reached and

$$\tau_\beta \leq \frac{\beta}{2\sigma(-\beta)}. \quad (61)$$

Finite-width outcome/learnability transfer follows from Theorem 16 only after deterministic witness margins and its common tube event are established; ordinary floating-point integration is not silently promoted to such a proof.

7 Counterfactual Drift Correction

Flatten the implemented trainable tensors into one fixed parameter chart with the standard product Euclidean/Frobenius inner product. Every gradient, dot product, norm, projection, Lipschitz ball, and straight segment in this section uses that chart and metric; no reparameterization-invariance or natural-gradient claim is made.

At a both-feature state θ , let

$$a = \nabla m_s(\theta), \quad p = \nabla m_w(\theta), \quad v = -\nabla L(\theta), \quad F_s = a \cdot v, \quad F_w = p \cdot v. \quad (62)$$

A same-initialization weak-only shadow supplies F_w^W . Define

$$q = \begin{cases} p - \frac{p \cdot a}{\|a\|^2} a, & a \neq 0, \\ p, & a = 0, \end{cases} \quad \delta = [F_w^W - F_w]_+, \quad \alpha = \begin{cases} \delta / \|q\|^2, & \delta > 0, q \neq 0, \\ 0, & \delta = 0. \end{cases} \quad (63)$$

The corrected velocity is $v' = v + \alpha q$. A positive deficit with $q = 0$ is infeasible. The theorems assume full-batch gradients, plain SGD, zero weight decay, and no clipping. A nonbinding coefficient cap leaves the uncapped result unchanged; a binding cap preserves orthogonality but generally loses target attainment, so no target-attaining optimality claim then applies.

Theorem 18 (CDC local guarantees). *The construction in Eq. (63) has the following fixed-coordinate properties.*

1. **Instantaneous preservation:** $a \cdot v' = a \cdot v$ exactly, including when α is capped.
2. **Minimum Euclidean norm:** for the feasible uncapped problem, $d^* = \alpha q$ is the unique minimum-norm, lower-bound-attaining correction satisfying $a \cdot d = 0$ and $p \cdot (v + d) \geq F_w^W$.
3. **Local finite-step bound:** if ∇m_s is L -Lipschitz in the same norm on a convex neighborhood containing both straight coordinate step segments, then

$$|m_s(\theta + \eta v') - m_s(\theta + \eta v)| \leq \frac{L}{2} \eta^2 (\|v'\|^2 + \|v\|^2). \quad (64)$$

For the active uncapped correction, if $\|v\| \leq V$, $\delta \leq D$, and $\|q\| \geq q_{\min} > 0$, the right side is at most

$$\frac{L}{2} \eta^2 [V^2 + (V + D/q_{\min})^2]. \quad (65)$$

This explicit norm bound also remains valid under a nonnegative upper cap, which can only reduce α .

For a positive raw deficit, exact feasibility can be stated intrinsically only as dm_w having a nonzero restriction to $\ker(dm_s)$; in this metric it is equivalent to $q \neq 0$. The represented q , its norm, the numerical `feasibility_epsilon`, and the selected minimum-norm correction are metric- and parameter-scale-dependent. The finite-step result is one-step and local in the same chart. It is not a trajectory guarantee and does not imply final strong-response retention.

8 Executable scope guards

Appendix Table 4 maps theorem objects to their implementations and explicit scope guards. Helpers enforce only represented scalar or configuration conditions; callers still establish trajectory, model, first-entry, and smoothness hypotheses. Unit tests check software identities and edge cases, not mathematical propositions or unrepresented assumptions.

Every general positive-disorder/positive-lag solver path that would require an unproved closure raises `NotImplementedError`. This is a scientific guard, not missing numerical polish.

9 Experimental protocol

9.1 Synthetic task and pairing

Sequences have two input channels. A strong cue of scale ρ and a unit weak cue appear at controlled sequence positions separated by a lag. Positive tasks align the cues with the label; the negative control reverses the strong cue. Data and initialization are paired across both-feature and weak-only conditions. Training is deterministic full-batch cross-entropy with zero weight decay. Except for E-NL, the two conditions can be run sequentially because deterministic full-batch updates preserve the same trajectories; E-NL uses lockstep models so both exact drifts are available at each logged time.

Appendix Table 5 records model, task, seed, and compute settings, including the roughly doubled per-step model work required by matched- shadow methods. E1 uses four strengths and seed IDs 4–11, so each lag contains 32 ρ -seed cells but only eight independent seed IDs. Its gain is 0.5, $\beta = 0.25$, and horizon is $H = 5$; the negative control is one eight-seed cell at $(\rho, \text{lag}) = (4, 4)$. E2 uses gain 0.5 and a held-out empirical closure reference. E-NL uses one cell ($\rho = 4$, lag 2), eight seeds per architecture, gain 0.8, cue noise 0.1, zero background noise, $\beta = 0.5$, and direct autograd drift. The frozen-kernel studies use the same task cell. E3 compares ERM, Spectral Decoupling, an interaction penalty, CDC, loss-gradient projection, unconstrained rescue, Bloop-style shadow-target rescue, and PCGrad-style shadow-target rescue; the five rescue arms share the same weak-only shadow target. These style labels do not assert canonical source parity.

Two prospective protocols are frozen but unexecuted. The semi-real design plans 64 paired records on MNIST 3-vs-8 and FashionMNIST 0-vs-6 with real core pixels and an exactly removable generated cue channel. The expanded design plans 192 nonlinear records over $\beta \in \{0.25, 0.5, 1, 2\}$ and 256 CDC-tradeoff method-records. Their validators both report that execution is not authorized; unresolved data, environment, commit, seed, telemetry, parity, and claim gates prevented scientific runs. No optimizer-backed outcome or empirical claim was generated. The excluded evaluator-engineering set and its non-scientific use are documented in Appendix C.

Manifest-sealed prediction protocol. `enl-preflight` constructs no optimizer and observes no trained state. It writes the initialization kernels and predictions, verifies state/gradient nonmutation, and hashes every declared file and per-record kernel together with the resolved config and executable-source fingerprint. Before constructing an optimizer, `enl-evaluate` requires the sidecar, actual manifest, and externally pinned digest to agree; verifies every hash/size, source, protocol, seed pair, and time grid; and loads rather than rebuilds predictions. It re-hashes all sealed inputs after training. The broad five-endpoint rule was frozen before its pilot; after that rule failed, only drift/response crossing was frozen as primary on fresh seeds. The latter crossed design uses a two-way pigeonhole bootstrap with 10,000 replicates.

Statistics and censoring. Independent seed pairs are the unit of replication only when neither seed axis is reused. E1’s 32 cells per lag reuse eight seed IDs across four strengths, and the E-NL 3/8, 5/8, and 0/8 proportions describe only the eight observed seeds per architecture; none is an estimate of universal event prevalence. Under E20’s crossed data/model-seed reuse, row-wise Student- t intervals are suppressed and the two-way percentile interval resamples the observed seed axes independently. It is conditional on those observed levels: an all-correct table necessarily gives $[1, 1]$ and does not quantify unseen failure modes. Hitting times beyond the horizon are right-censored and never converted into finite delays. Conditional crossing-time means are reported only with their crossing-record denominator. The exploratory E-NL 5×5 grid uses seeds 0–3 and is explicitly post-hoc; it is not used to select a confirmatory condition.

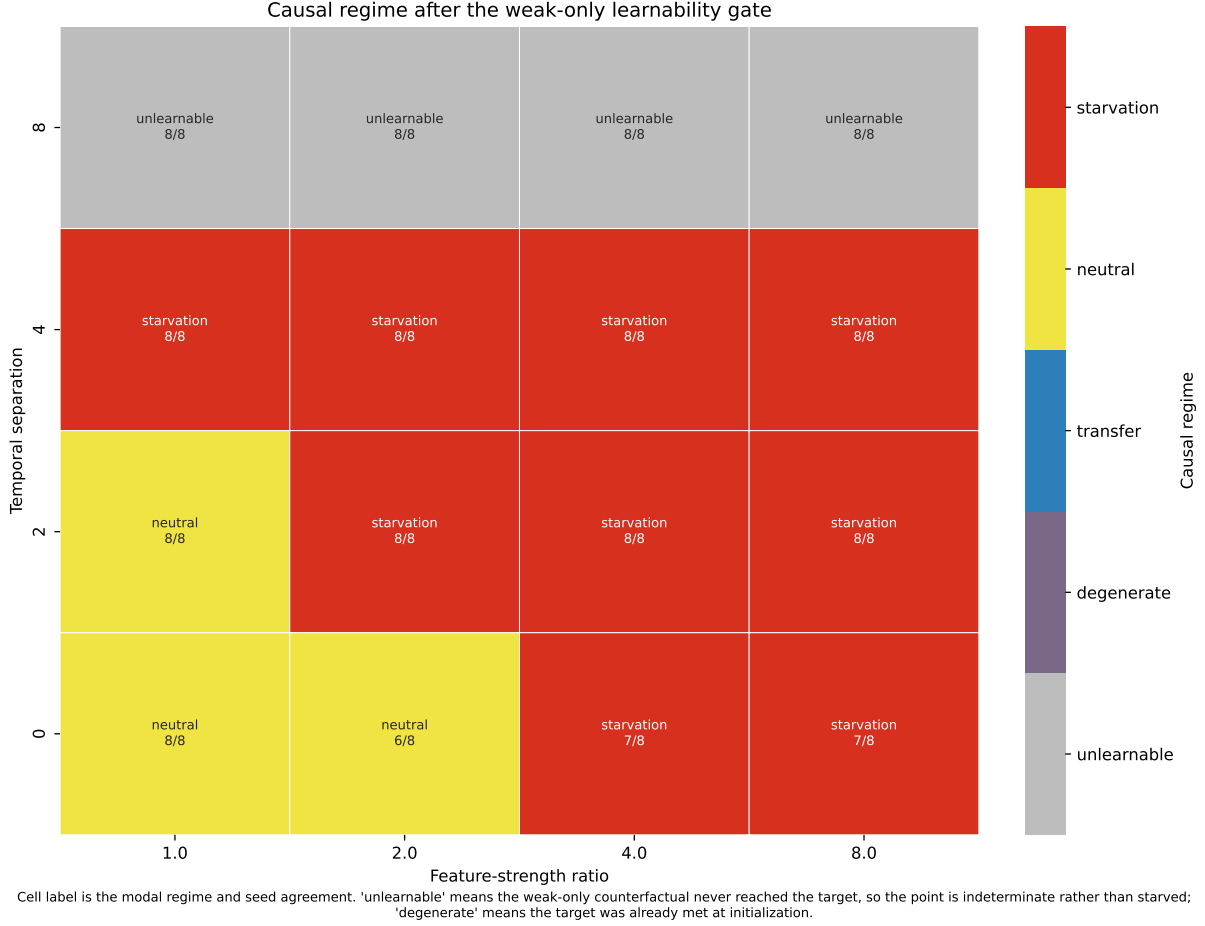


Figure 1: Controlled dense-linear causal regions. The learnability gate prevents lag-8 points from being mislabeled as starvation.

Reproducibility. Ordinary runs write resolved configuration, environment, trajectory, summary, and figure artifacts; frozen studies additionally seal predictions and hashes before training. The final E-NL run archives a 17-file executable-source fingerprint and unmodified metadata, making that named run reconstructible. E19 and E20 are different: their manifests, predictions, scores, seals, and historical source fingerprints survive, but the exact dirty Python source bytes were not archived and are unavailable and unreconstructible. Current source intentionally fails their source guard; a new preflight/evaluation is a new study, not a replay. Historical E1, E2, E2-R, exploratory E-NL, and E3 artifacts likewise disclose dirty states predating source fingerprinting.

10 Results

10.1 Controlled dense-linear phase grid

At lags 0, 2, and 4, all 32 ρ -seed cells per lag (four strengths by eight seed IDs) have positive weak AUC gaps. Averaged across ρ , the lag-wise gaps are 5.76, 8.93, and 6.11, respectively; the effect peaks at lag 2 rather than decreasing monotonically with lag. At every fixed lag, the gap increases monotonically over the measured ρ grid. The negative-control cell is transfer in 8/8 seeds, with mean gap -4.038 and 95% CI $[-4.498, -3.578]$. Every lag-8 cell is weak-only unlearnable, so these 32 rows are not labeled starvation. Final accuracy is 1.0 throughout, and no row meets the target at initialization.

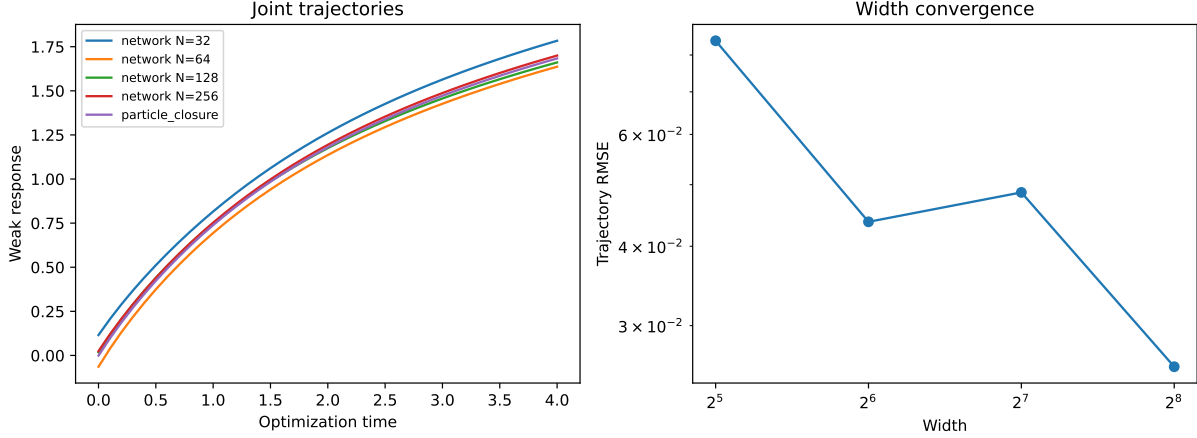


Figure 2: Finite-width agreement with a held-out empirical finite-network closure reference. This is not DMFT validation.

Table 1: Final E-NL result at $\rho = 4$, lag 2, eight seeds. Crossing times and confidence intervals are conditional on crossing and must not be read as eight-seed estimates.

Model	exact drift	response	causal	mean τ_d	mean τ_r	max $ R_w^B - R_w^W $
tanh	3/8	3/8	3/8	0.194	0.389	0.02197
GRU	5/8	5/8	0/8	3.846	7.200	0.002986

10.2 Empirical width trends and exact singular checks

E2 compares finite networks with a disjoint-seed empirical *closure reference*; it is not an independently derived theory. Overall both-condition trajectory NRMSE is 0.09084, 0.04761, 0.05149, 0.02958 at widths 32, 64, 128, 256. Weak-only NRMSE is 1.56931, 0.38792, 0.40201, 0.20024. Both improve end-to-end but are non-monotone at $N = 128$. All 32 point-condition-diagnostic combinations improve end-to-end; large normalized weak-only values at difficult points have near-zero denominators. The exact finite-width geometry check has zero recorded relative error and the maximum projected-identity error is 2.86×10^{-6} .

In the exact zero-disorder solver check, the time-discretization residual scales as $O(\eta)$ with fitted slope 1.0115. A frozen-geometry integrator agrees with the projected-flow integrator to 1.91×10^{-15} . The general solver acceptance record remains false by construction because all general-DMFT checks are blocked.

10.3 Nonlinear crossover is seed-dependent

Table 1 is the authoritative nonlinear result. Tanh has a complete exact-drift/response/tail-area causal certificate in 3/8 seeds; the other five have negative exact relative drift from initialization and therefore do not exhibit transfer-to-starvation. GRU has exact drift and response crossings in 5/8 seeds, but the weak-only trajectory fails to reach $\beta = 0.5$ in all eight seeds; its causal count is therefore zero. The nonzero direct-versus-projected residuals empirically confirm that Gg cannot be silently treated as the exact nonlinear probe derivative.

The preregistered gate is the first hit at $\beta = 0.5$, but a single binary operating point can hide whether failures are marginal. We therefore add a post-hoc terminal audit using $B_{W,s}(H) = M_{w,s}^W(H)$ and the empirical fraction of seeds with $B_{W,s}(H) \geq \beta$ over candidate thresholds. No model was re-trained: Figure 3 is derived from the tracked per-seed E-NL summary. Tanh terminal min/mean/max are 4.41313266754/4.49664855003/4.67605876923, with 8/8 above $\beta = 0.5$; GRU min/mean/max are 0.0828229486942/0.129690139554/0.187142759562, with 0/8 above 0.5. Thus the GRU failure is not marginal at that operating point. Terminal reach is not generally interchangeable with the preregistered first-hit event on a nonmonotone trajectory, and this post-hoc diagnostic supports neither a new confirmatory endpoint nor

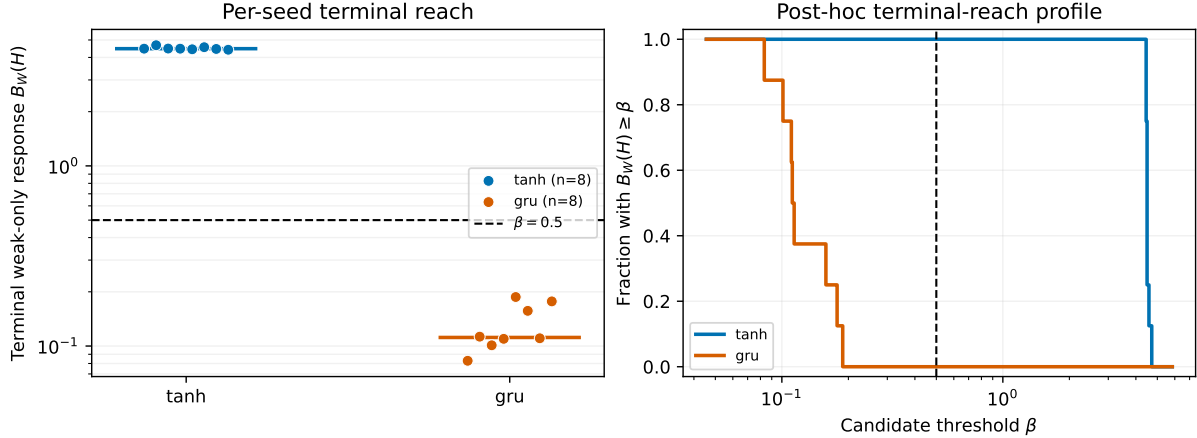


Figure 3: Post-hoc weak-only reach audit from the completed E-NL run. Points are terminal responses for all eight seeds; horizontal segments mark medians and the dashed line marks the preregistered $\beta = 0.5$. The right panel sweeps candidate thresholds without rerunning training. Terminal reach is not generally interchangeable with a first-hit gate for nonmonotone trajectories.

Table 2: Manifest-sealed frozen-kernel evaluations. The broad pilot’s complete five-output rule failed. Only the two event rows in the fresh factorial were preregistered primary endpoints. CIs resample data- and model-seed axes.

Protocol	Endpoint	Overall	tanh	GRU	95% CI
Broad pilot	phase	12/16	6/8	6/8	diagnostic
Broad pilot	weak learnability	12/16	4/8	8/8	failed
Restricted 4×8	drift event	60/64	32/32	28/32	[0.8125, 1]
Restricted 4×8	response event	64/64	32/32	32/32	[1, 1]

an architecture ranking.

The post-hoc tanh search over $\rho \in \{1.5, 2, 3, 4, 6\}$ and lags $\{0, 1, 2, 4, 8\}$ found no cell with 4/4 causal certificates; the best cells reached 3/4. We retain this as a negative search audit and do not select a searched cell for a paper claim. In particular, the superseded 8/8 tanh result and its $\tau^* = 1.6112$ used a different response convention and are not valid under the final definition.

10.4 E19 fails broadly; E20 succeeds only on restricted event labels

The initialization-frozen tangent surrogate was first tested under a sealed five-output pilot using one unseen data seed and eight unseen model seeds per architecture. It obtained 12/16 phase, 14/16 drift-crossing, 14/16 response-crossing, 14/16 causal- certificate, and 12/16 weak-learnability classifications. The preregistered rule nevertheless *failed*: tanh learnability was 4/8, below the 5/8 per-model floor. We preserve that rejection rather than averaging it away.

Only after this failure did we define the narrower crossing-event hypothesis. A fresh sealed factorial crossed four unseen data seeds with eight unseen model seeds for each architecture. Table 2 reports classification correctness, not observed crossing prevalence. The restricted rule passed: all response-crossing labels and 60/64 drift-crossing labels were correct; the two-way bootstrap lower endpoints exceeded the frozen 0.5 floor.

The restriction is substantive. In the factorial, phase, causal-certificate, and learnability correctness were only 52/64, 56/64, and 49/64 and were explicitly secondary; tanh learnability was 17/32. Crossing times remained systematically early (overall drift/response bias $-0.822/ -1.189$), and tanh response-gap and weak- response RMSE were 2.455 and 2.489 despite high sign agreement. Thus the evidence supports an uncalibrated event classifier at this one width/task cell. It neither proves Theorem 5’s kernel premise nor validates phase, causal starvation, learnability, trajectory magnitude, calibrated time, or prevalence.

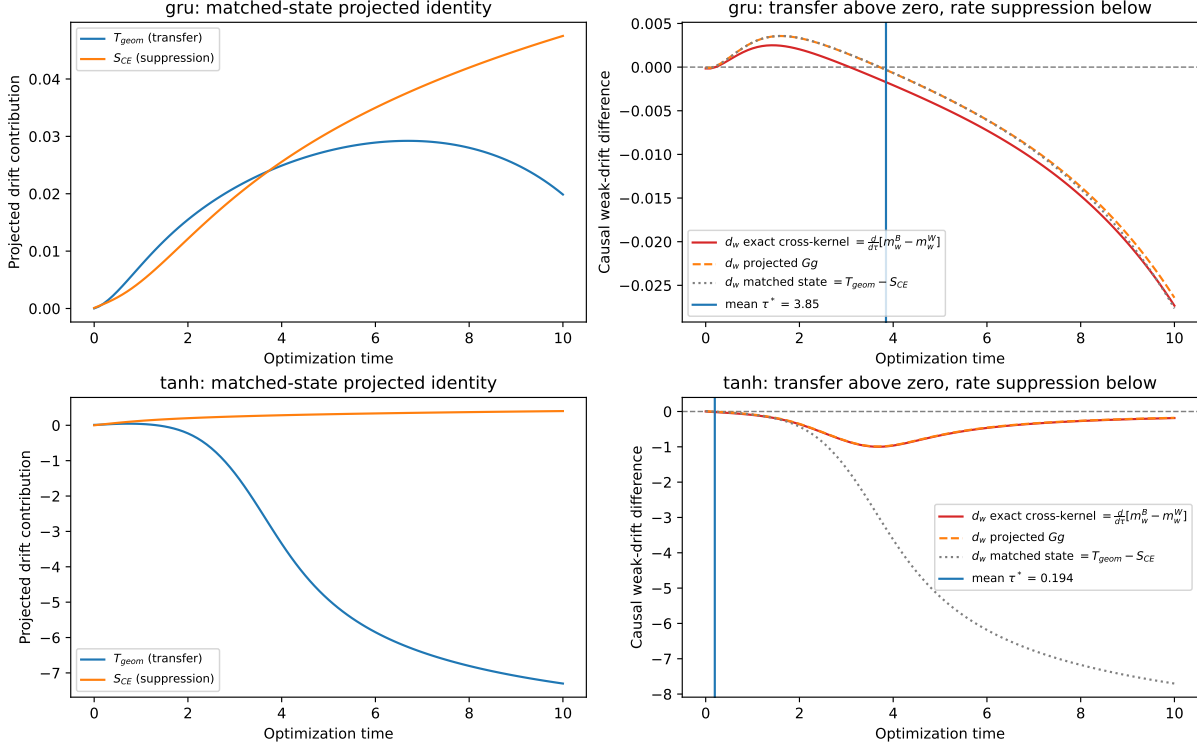


Figure 4: Theorem-aligned nonlinear trajectories. Direct-autograd drift is primary; projected and matched-state quantities are separately styled diagnostics.

Appendix D reports the complete failure, restricted-success, timing-bias, and magnitude-error numbers.

10.5 CDC has the local guarantee, not the best final tradeoff

All five shadow-based methods reduce the causal weak AUC gap from ERM’s 10.5155 to approximately zero and attain their shared weak-drift target at every logged step. Their weak outcomes are numerically similar to roughly four decimal places, but not statistically equivalent: 9/10 direct paired comparisons are significant at 0.05, while the differences are practically tiny relative to the roughly 10.5 change from ERM. CDC alone preserves instantaneous strong drift to rounding scale, 4.77×10^{-7} , versus 1.48×10^{-1} to 3.75×10^{-1} for the alternative shadow constraints. This local property does not yield the best final strong response: unconstrained Rescue and the PCGrad-style arm end at 1.295, compared with CDC’s 1.042 and ERM’s 1.364.

11 Discussion and limitations

Theory and causal scope. The finite-width cross-kernel identity, exact sample-logit/response dynamics, initialization-frozen tangent surrogate, conditional continuous/secant-SGD comparison, dense-linear geometry, cumulative drift balance and single-crossing corollary, conditional rank-one criterion, transverse hitting-time bound, singular wide limit, and fixed-chart local CDC results are mathematical statements with proofs. Their scopes differ. The cumulative balance characterizes outcome suppression but does not supply weak-only learnability; the rank-one criterion addresses rate crossing; the frozen comparison assumes unproved kernel-movement control; the wide limit is restricted to zero disorder and zero lag; and CDC has a local one-step guarantee, not a trajectory theorem.

Table 3: Dense CDC ablation, eight paired seeds. Shadow-based arms have access to a matched weak-only model; the other arms do not. “Strong change” is the maximum absolute instantaneous strong-drift perturbation.

Method	Shadow	Weak AUC gap	Strong change	Final m_s
ERM	no	10.5155	—	1.364
Spectral Decoupling	no	9.0229	—	0.829
Interaction	no	10.5145	—	1.365
CDC	yes	−0.0005	4.77×10^{-7}	1.042
Loss-gradient projection	yes	−0.0002	3.75×10^{-1}	0.371
Unconstrained rescue	yes	−0.0007	1.48×10^{-1}	1.295
Bloop-style	yes	0.0002	3.75×10^{-1}	0.370
PCGrad-style	yes	−0.0007	1.48×10^{-1}	1.295

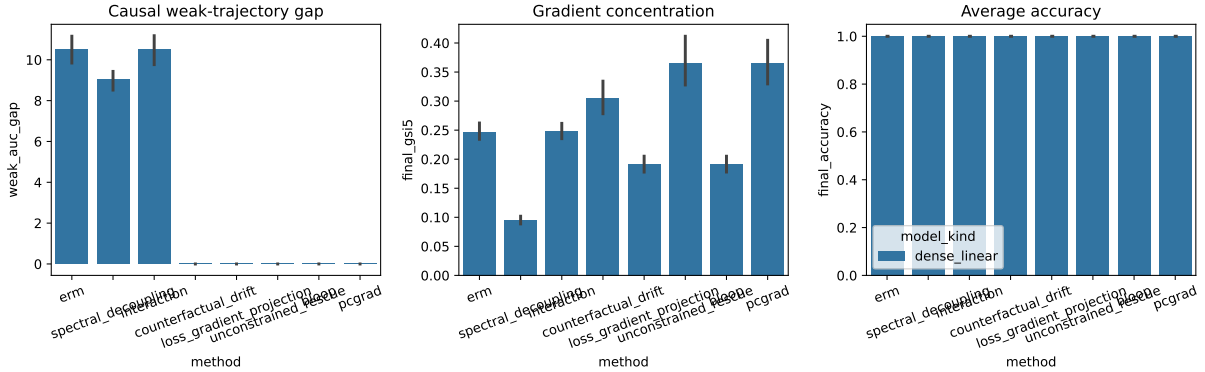


Figure 5: Eight-method dense-linear mitigation ablation. CDC’s established distinction is instantaneous first-order preservation, not superior weak rescue or final strong retention.

Empirical interpretation. E1 shows positive weak AUC gaps in all 32 ρ -seed cells per lag at lags 0, 2, and 4, but those cells reuse eight seed IDs. E-NL contains only eight seeds per architecture at one synthetic task cell: tanh is seed-dependent, and GRU exhibits outcome crossings without passing the causal gate. The post-hoc terminal profile shows that the GRU failure is not marginal at $\beta = 0.5$, but it neither replaces the first-hit gate nor ranks architectures. These counts are descriptive, not prevalence estimates. The exploratory search found no 4/4 robust cell.

Frozen-surrogate sequence. E19 failed its preregistered five-endpoint rule; correctness on other labels cannot rescue the tanh learnability failure. E20 was designed only after that rejection and supports a narrower claim: drift/response crossing-event classification on one fresh crossed-seed factorial. Its phase, causal, and learnability labels were secondary, its time predictions were systematically early, and tanh trajectory magnitudes were poor. Manifest integrity proves that frozen inputs were consumed unchanged; it does not prove kernel stability or surrogate accuracy.

CDC and baselines. CDC uniquely preserves the instantaneous strong drift within the tested shadow-method family, as its theorem predicts. The ablation does not show a general practical advantage: weak rescue is numerically similar across shadow arms, and other arms retain a larger final strong response. CDC also assumes a constructible weak-only shadow and roughly doubled model work. The Bloop-style and PCGrad-style arms are repository implementations motivated by those methods, not independently audited replicas.

Open theory and external validity. A positive-disorder or positive-lag optimization-time DMFT still requires an effective process, visible and loss-invisible closure variables, finite-horizon well-posedness, joint quenched concentration, and discretization/hitting-time control. Neither an empirical closure nor event classification supplies these objects. An analytic ρ_c and certified or quantitatively accurate tanh/GRU crossing times therefore remain open. All citable outcomes remain synthetic; no real-data or semi-real experiment has been run. The frozen generated-cue MNIST and FashionMNIST intervention is defensible at the design level but unauthorized and outcome-free, and the expanded beta/nonlinear/CDC protocol is likewise unexecuted. Neither changes the empirical scope. The unrun Waterbirds adapter is a minibatch head-coordinate surrogate outside the full-batch matched-shadow theorem, and its intercept is not an identified bird-shape response. We make no Waterbirds claim.

Provenance. The final E-NL source is reconstructible through its archived 17-file executable- source fingerprint and preserved metadata. E19 and E20 preserve compact outcomes, seals, and source digests, but not the exact dirty source bytes; they are auditable but not historically replayable. Earlier E1, E2, E2-R, exploratory E-NL, and E3 runs share a disclosed dirty-source limitation.

12 Conclusion

Gradient starvation requires more than a declining weak-feature rate. Relative rate suppression becomes outcome suppression only when cumulative negative drift exceeds the earlier positive advantage; causal starvation additionally requires that the matched weak-only trajectory first-hit the learnability target. Exact finite-width response dynamics and the general cumulative balance make these separations formal. The realized-initialization threshold is exact, but its positive-lag sign is seed-dependent and its trajectory consequences require independently proved product-ball constants. The rank-one, hitting-time, singular wide-limit, and CDC results state their assumptions explicitly; the singular reference starts suppressed and does not supply a transfer phase.

The empirical sequence supports the same discipline. Corrected E1 is positive across 32 ρ -seed cells per lag, not 32 independent seeds. In E-NL, tanh causal crossover is seed-dependent and GRU outcome crossings fail the learnability gate; the post-hoc terminal profile strengthens that operating-point diagnosis without replacing the gate. E19 remains a failed broad predictor, whereas E20 supports only restricted drift/response crossing-event labels. CDC demonstrates its exact local constraint but not broad practical superiority. With eight-seed nonlinear evidence, no real-data experiment, unreconstructible historical dirty source for E19/E20, and no positive-disorder/positive-lag DMFT, general recurrent causal prediction and calibrated crossover times remain open. The frozen semi-real and expanded nonlinear/CDC protocols remain prospective design artifacts with no outcomes, so they do not alter that conclusion.

A Proofs for the finite-width theorem package

A.1 Proof of Proposition 1 and Corollary 2

For $\ell(r) = \log(1 + e^{-r})$, $\ell'(r) = -\sigma(-r)$. Therefore

$$\dot{\theta}_C = -\nabla L_C = \frac{1}{n_C} \sum_i \sigma(-r_i^C) \nabla r_i^C. \quad (66)$$

The chain rule gives

$$\dot{M}_a^C = \langle \nabla M_a^C, \dot{\theta}_C \rangle = \frac{1}{n_C} \sum_i \langle \nabla M_a^C, \nabla r_i^C \rangle \sigma(-r_i^C), \quad (67)$$

which proves Proposition 1. Under exact realization, $\nabla r_i^C = \sum_b z_{ib}^C \nabla M_b^C$. Substitution and exchange of finite sums yield Eq. (7). If an orthogonal residual logit term is present, substitution of $\nabla r_i^C = \sum_b z_{ib}^C \nabla M_b^C + \nabla r_i^\perp$ instead yields Eq. (8). $\square$

A.2 Proof of Corollary 3 and Theorem 5

The parameter flow above can be written $\dot{\theta} = J^\top q(r)/n$. Therefore the chain rule gives $\dot{r} = J\dot{\theta} = JJ^\top q(r)/n$ and $\dot{M} = \langle \nabla M, \dot{\theta} \rangle = c^\top q(r)/n$, proving Corollary 3.

For the comparison, q is 1/4-Lipschitz componentwise and hence in Euclidean norm. Subtracting Eq. (11) from Eq. (10), adding and subtracting $\Theta_0 q(r)$, and writing $e_r = \|r - \bar{r}\|$ gives

$$e'_r(u) \leq \frac{\|\Theta_0\|}{4n} e_r(u) + \frac{B_\Theta V}{n} = a e_r(u) + \frac{B_\Theta V}{n}. \quad (68)$$

Grönwall with $e_r(0) = 0$ proves Eq. (13). The same add-subtract step gives

$$|\dot{M} - \dot{\bar{M}}| \leq \frac{B_c V}{n} + \frac{\|c_0\|}{4n} e_r. \quad (69)$$

Integrating and using $\int_0^t \phi_a(u) du = \psi_a(t)$ proves Eq. (14).

For SGD, the fundamental theorem of calculus along the straight update segment gives

$$r_{j+1} - r_j = \tilde{J}_j \delta \theta_j = \frac{\eta}{n} \tilde{\Theta}_j q(r_j), \quad (70)$$

$$M_{j+1} - M_j = \langle \tilde{m}_j, \delta \theta_j \rangle = \frac{\eta}{n} \tilde{c}_j^\top q(r_j). \quad (71)$$

Subtracting the frozen Euler recursion yields scalar recurrences

$$e_{j+1} \leq (1 + \eta a) e_j + \frac{\eta D_\Theta V}{n}, \quad (72)$$

$$E_{j+1} \leq E_j + \frac{\eta D_c V}{n} + \frac{\eta \|c_0\|}{4n} e_j. \quad (73)$$

Solving the first geometric recurrence and summing it in the second gives Eq. (18), including the stated $a = 0$ limits. Paired bounds are the triangle inequality. Error-separated sign transfer follows immediately; first-hit transfer requires the additional assumptions of Corollary 14. $\square$

A.3 Proof of the dense-linear geometry

For a unit response mode in channel a at lag ℓ_a , write b_a for the corresponding input column. Direct differentiation of $c^\top W^{\ell_a} b_a$ gives gradient $p_a = W^{\ell_a} b_a$ with respect to c , gradient $q_a = (W^{\ell_a})^\top c$ with respect to column b_a , and recurrent gradient

$$S_a = \sum_{j=0}^{\ell_a-1} (W^j)^\top c (W^{\ell_a-1-j} b_a)^\top. \quad (74)$$

The parameter blocks are disjoint, so their gradient inner products add. Input columns contribute only when the channels coincide, giving Eq. (20). $\square$

A.4 Proof of Theorem 6

For the positive rank-one cue laws, $g^B = s_B(\rho, 1)$ and $g^W = s_W(0, 1)$ at the shared initialization. Corollary 2 and the exact dense geometry give

$$\dot{M}_w^B(0) = s_B(\rho G_{ws} + G_{ww}), \quad \dot{M}_w^W(0) = s_W G_{ww}. \quad (75)$$

Subtraction, with $K = G_{ws}$ and $A = G_{ww}$, proves Eq. (26). Since $\rho s_B > 0$, rearrangement gives Eq. (27). The displayed K, A, S_w formulas are Eq. (20) specialized to a zero-lag strong mode and an ℓ -lag weak mode. Under the iid width scaling at fixed positive ℓ , expanding the gates at zero gives

$$d_0 = \rho \left(\frac{K}{2} - \frac{A M_s(0)}{4} \right) + O_{\mathbb{P}}(N^{-1}), \quad (76)$$

whose leading sign is centered and realization-dependent. This proves the exact criterion and the stated obstruction to a distribution-only positive-lag boundary. $\square$

A.5 Proof of Proposition 7 and Corollaries 8–9

The positive and negative parts satisfy $d = d_+ - d_-$ almost everywhere. Absolute continuity and the fundamental theorem of calculus therefore give

$$\Delta(t) = \Delta(0) + \int_0^t d(u) du = P(t) - N(t), \quad (77)$$

proving Proposition 7 and its pointwise sign equivalences. Under Corollary 8's sign pattern, $d_- = 0$ before τ_d and $d_+ = 0$ afterward. Hence $P(t) = P(\tau_d) = A_+$ and

$$\Delta(t) = A_+ - N(t), \quad t \geq \tau_d. \quad (78)$$

The post-crossing sign assumption makes N continuous and strictly increasing on every nontrivial post-crossing interval. Later equality, strict negativity, and uniqueness are therefore respectively equivalent to $N(H) \geq A_+$, $N(H) > A_+$, and strict post-crossing growth of N .

If $d \leq -\kappa$ after t_0 , then $N(t) - N(t_0) \geq \kappa(t - t_0)$; solve for the time needed to cover the remaining area. If $d'(t) \leq -\lambda$ and $d(\tau_d) = 0$, integration gives $d(t) \leq -\lambda(t - \tau_d)$ and then $N(t) - N(\tau_d) \geq \lambda(t - \tau_d)^2/2$. Since $N(\tau_d) = 0$ under the single-crossing hypotheses, applying Corollary 8 proves both stated bounds. $\square$

A.6 Proof of Theorem 10 and Corollary 11

Before exit from U_R , path length is at most $S_R t$, so exit cannot occur before $h_R = R/S_R$. Along this interval, $d' = j$, and integration of Eq. (32) gives

$$d_0 - \Lambda t \leq d(t) \leq d_0 - \lambda t. \quad (79)$$

The lower bound is positive before d_0/Λ , the upper bound is negative after d_0/λ , continuity gives a zero between them, and $d' < 0$ makes it unique. This proves Eq. (33). Integrating the upper bound from $\Delta(0) = 0$ gives

$$\Delta(t) \leq d_0 t - \frac{\lambda t^2}{2} < 0 \quad \text{for } t > 2d_0/\lambda. \quad (80)$$

The weak-only lower drift bound reaches β by the time in Eq. (34), proving the causal conclusion.

For the safe case, $|j| \leq J$ gives $d(t) \geq d_0 - Jt$; one more integration gives Eq. (35). The strict premise $d_0 - Jh_R > 0$ makes both quantities positive on the certified horizon. $\square$

A.7 Proof of Theorem 12 and Corollary 13

The deterministic cue law gives $g^B = s_B(\rho, 1)$ and $g^W = s_W(0, 1)$. Equation (7) therefore gives $F_w^B = s_B H_B$, $F_w^W = s_W H_W$, $u'_B = s_B Q_B$, and $u'_W = s_W H_W$. Since

$$\frac{d}{du} \log \sigma(-u) = -(1 - \sigma(-u)), \quad (81)$$

differentiating Ψ yields Eq. (41). When both drifts are positive, monotonicity of the logarithm gives $\text{sign}(F_w^B - F_w^W) = \text{sign}(\Psi)$.

Under $\Psi' \leq -\kappa$, integration gives $\Psi(t) \leq \Psi(t_0) - \kappa(t - t_0)$. Continuity forces a zero by $t_0 + \Psi(t_0)/\kappa \leq H$, while strict decrease gives uniqueness. The primitive inequalities imply the derivative bound term by term. $\square$

A.8 Proof of Corollary 14

Continuity and first-entry semantics imply $x(\tau) = \beta$. Absolute continuity and $x' \geq \kappa$ almost everywhere give, by integration,

$$x(t) \leq \beta - \kappa(\tau - t) \quad (\tau - r \leq t \leq \tau), \quad x(t) \geq \beta + \kappa(t - \tau) \quad (\tau \leq t \leq \tau + r). \quad (82)$$

On $[0, \tau - r]$, the quantitative prehistory margin and $\epsilon < \eta$ give $x_N < \beta$. On $[\tau - r, \tau - \epsilon/\kappa]$, the left cone and the uniform error again give $x_N < \beta$. Hence $\tau_N \geq \tau - \epsilon/\kappa$. At $t = \tau + \epsilon/\kappa$, the right cone gives $x_N(t) \geq \beta$, so the hit is finite and $\tau_N \leq \tau + \epsilon/\kappa$. This proves Eq. (46).

For sampled data, the first grid point at or after $\tau + \epsilon/\kappa$ is at most h later and, by the radius condition, remains inside the right cone; it is therefore at or above β . No sampled hit can precede the continuous lower bound. If the piecewise-linear interpolant first crosses on $[t_{j-1}, t_j]$, then $x_N(t_j) \geq \beta$, so $t_j \geq \tau - \epsilon/\kappa$ and $t_{j-1} \geq \tau - \epsilon/\kappa - h$; its upper bound is no later than the sampled hit. This proves Eq. (47). Applying the bound separately to two finite hits and using the triangle inequality proves the paired statement. $\square$

A.9 Proof of Theorems 15 and 16

At zero recurrent gain and lag, gradient flow gives $\dot{b}_a = g_a c$ and $\dot{c} = \sum_b g_b b_b$. Differentiating $M_a = c \cdot b_a$, $u_{ab} = b_a \cdot b_b$, and $w = c \cdot c$ proves Eq. (50). The mode gradient has one component in the c block and one in its own b_a block, yielding $G_{ab} = u_{ab} + \delta_{ab} w$.

At initialization, the variances of M_s, M_w, u_{sw} are $1/N$ and those of $u_{ss} - 1, u_{ww} - 1, w - 1$ are $2/N$. Coordinatewise Chebyshev bounds and a union bound sum to $1 + 1 + 1 + 2 + 2 + 2 = 9$, proving Eq. (52).

The field bound $\|g(M)\|_\infty \leq A$ follows from $0 \leq \sigma \leq 1$. Since σ is $1/4$ -Lipschitz,

$$|g_a(M) - g_a(\widetilde{M})| \leq \frac{1}{4} \sum_b \mathbb{E}|z_a z_b| \left\| M - \widetilde{M} \right\|_\infty, \quad (83)$$

which proves Eq. (54). Define the Gram energy $R = w + u_{ss} + u_{ww}$. Along the deterministic flow,

$$R' = 4g \cdot M = 4\mathbb{E}[X\sigma(-X)], \quad X = z \cdot M. \quad (84)$$

Because $x\sigma(-x) \leq c_\star = W(e^{-1})$ and $R(0) = 3$, $R(t) \leq R_H$. Every coordinate of q is a squared norm or inner product of b_s, b_w, c , so Cauchy-Schwarz gives $\|q(t)\|_\infty \leq R(t) \leq R_H$.

Consider two states in the cube $\|q\|_\infty, \|\tilde{q}\|_\infty \leq B_{H,r}$ and set $d = \|q - \tilde{q}\|_\infty$. Every product in the M and u equations satisfies the direct estimate

$$|g_a(M)y - g_a(\widetilde{M})\tilde{y}| \leq (A + B_{H,r}C)d. \quad (85)$$

There are at most three such products in an M equation and two in a u equation; the factor two and two cue indices give four in the w equation. Thus the full vector field is $L_{H,r} = 4(A + B_{H,r}C)$ -Lipschitz on the cube and therefore on the radius- r reference tube.

Let T_r be the first time before H that $\|q_N(t) - q(t)\|_\infty$ reaches the tube boundary. Up to T_r , Gronwall's inequality and Eq. (56) give

$$\|q_N(t) - q(t)\|_\infty \leq e^{L_{H,r}t} \|q_N(0) - q_0\|_\infty. \quad (86)$$

If $L_{H,r} > 0$, the initialization event makes this bound strictly less than r for every $t < H$, so the boundary cannot be reached before H . If $L_{H,r} = 0$, then $A = C = 0$, the cue law is zero almost surely, and the six-scalar vector field vanishes; the initial discrepancy is constant and cannot exit the closed tube. The endpoint may lie on the boundary. In either case the same estimate is at most δ throughout $[0, H]$, proving Eq. (57). Applying Eq. (52) with $\epsilon = e^{-L_{H,r}H} \min\{r, \delta\}$ proves Eq. (58). For the shared-initialization pair, use a common r and the maximum deterministic constant from the two fixed cue laws; the same concentrating initial six scalars control both paths. $\square$

A.10 Proof of Corollary 17

The six-scalar equations give equal initial weak drifts in the two conditions. Differentiating those drifts once at the symmetric state gives

$$(\dot{M}_w^B)'(0) = -\frac{\rho^2 + 1}{2}, \quad (\dot{M}_w^W)'(0) = -\frac{1}{2}, \quad (87)$$

so $\bar{d}'(0) = -\rho^2/2$. Taylor's theorem and continuity then make $\bar{\Delta}(t) < 0$ for all sufficiently small positive t .

The weak-only symmetric subsystem preserves $u_{ww} = w$. Direct differentiation shows $u_{ww}^2 - M_w^2 = 1$, yielding $u_{ww} = w = \sqrt{1 + M_w^2}$ and the displayed positive drift. While $0 \leq M_w \leq \beta$, it obeys $\dot{M}_w \geq 2\sigma(-\beta)$; integration proves Eq. (61). The finite-width statement is then exactly Theorem 16 applied at deterministic suppression and target-attainment witness times with margins smaller than both strict witness margins. $\square$

B Proofs for Counterfactual Drift Correction

B.1 Instantaneous preservation

If $a = 0$, both strong drifts vanish. Otherwise,

$$a \cdot q = a \cdot p - \frac{p \cdot a}{\|a\|^2}(a \cdot a) = a \cdot p - p \cdot a = 0, \quad (88)$$

so $a \cdot v' = a \cdot v + \alpha a \cdot q = a \cdot v$. This argument is independent of α , hence survives a cap. $\square$

B.2 Unique minimum-norm correction

Let $\bar{\delta} = F_w^W - F_w$ be the raw deficit and $\delta = [\bar{\delta}]_+$. If $\bar{\delta} \leq 0$, $d = 0$ is feasible and uniquely norm-minimal. Suppose $\bar{\delta} > 0$, so $\delta = \bar{\delta}$. Any feasible d obeys $a \cdot d = 0$ and $p \cdot d \geq \delta$. Because $p = q + (p \cdot a / \|a\|^2)a$ when $a \neq 0$ (and $q = p$ when $a = 0$), $p \cdot d = q \cdot d$. Cauchy–Schwarz in the fixed Euclidean subspace orthogonal to a gives

$$\delta \leq q \cdot d \leq \|q\| \|d\|, \quad \|d\| \geq \delta / \|q\|. \quad (89)$$

Equality requires a nonnegative multiple of q . The correction $d^* = (\delta / \|q\|^2)q$ is feasible and attains this lower bound, so it is the unique minimum-norm, lower-bound-attaining feasible correction. Larger positive multiples can also attain the target but have larger norm. A positive deficit is feasible exactly when $q \neq 0$ in this representation. A nonbinding cap leaves d^* unchanged; a binding cap generally loses target attainment, so the target-attaining optimization claim no longer applies. $\square$

B.3 Local finite-step deviation

In the fixed chart and norm, L -Lipschitzness of the gradient on both straight segments gives the smoothness remainder

$$|m_s(\theta + h) - m_s(\theta) - \nabla m_s(\theta) \cdot h| \leq \frac{L}{2} \|h\|^2. \quad (90)$$

Apply this at $h = \eta v$ and $h = \eta v'$. Instantaneous preservation cancels the linear terms and the triangle inequality yields Eq. (64) for the actual velocities, capped or uncapped. For active uncapped CDC,

$$\|v'\| \leq \|v\| + \frac{\delta}{\|q\|} \leq V + D/q_{\min}, \quad (91)$$

which gives the explicit bound; a nonnegative upper cap can only decrease the added multiple of q . If smoothness is known only on the fixed-coordinate ball $B(\theta, r)$, sufficient segment conditions are $\eta V \leq r$ and $\eta(V + D/q_{\min}) \leq r$. These quantities and L are chart- and metric-dependent. $\square$

C Discrete certificate and experiment audit

For logged values Δ_k , increments telescope exactly:

$$\Delta_K - \Delta_0 = \sum_{k=0}^{K-1} (\Delta_{k+1} - \Delta_k). \quad (92)$$

The executable certificate requires one strict block of positive increments followed by one strict block of negative increments. The peak-minus-initial value is its positive area; peak-minus-final is its negative tail area. It certifies causal starvation only if the final trajectory is strictly below the initial gap and the external weak-only learnability flag is true. This finite-step certificate makes no assertion between logged points.

Excluded engineering outcomes. Data seed 100 with model seeds 1000–1007 is excluded from scientific evidence. A first frozen-prediction grid mismatch stopped before training; a second observed- trajectory grid mismatch was detected after nonlinear outcomes had been generated, but no scores were inspected. The set was used only to repair evaluator alignment through integer-step matching plus a numerical optimization-time check. It did not change labels, thresholds, or the five E19 endpoints; E19 and E20 use fresh seeds.

Table 4: Executable contracts in `src/gradient_starvation/theory.py`.

Contract	Mathematical object	Scope guard
<code>projected_statistics</code>	direct Eq. (6), Gg , residual	direct drift primary for nonlinear probes
<code>initial_frozen_empirical_kernel</code>	J_0, Θ_0, c_0 in Corollary 3	initialization only; full sample kernel
<code>integrate_frozen_logistic_sgd</code>	explicit Euler for Eq. (11)	exact for tangent surrogate, not nonlinear model
<code>frozen_kernel_discrete_error_bound</code>	Eq. (18)	caller supplies secant-kernel movement bounds
<code>equal_time_drift_difference</code>	Δ' and two projected product orderings	exposes non-canonical attribution
<code>dense_linear_initial_gap_certificate</code>	Eqs. (26)–(27)	noiseless positive rank-one dense-linear scope only
<code>parameter_tube_certificate</code>	Theorem 10, Corollary 11	caller supplies uniform product-ball constants; missing premises fail closed
<code>rank_one_drift_ratio</code>	Theorem 12 factors and Ψ	caller supplies noiseless/exact hypotheses
<code>discrete_crossover_certificate</code>	telescoping finite-step tail areas	learnability gate required for causal label
<code>transverse_hitting_time_error_bound</code>	$\epsilon/\kappa + h$	supplied error, prehistory, radius, and mesh guards
<code>zero_disorder_failure_bound</code>	Eqs. (52), (58)	caller supplies reference-tube constants
<code>cdc_finite_step_deviation_bound</code>	Eq. (64)	caller supplies local smoothness and segment containment

Table 5: Final synthetic experiment configurations. “Shadow” methods train a second weak-only model and therefore use roughly twice the model memory and per-step compute.

Experiment	Model / width	Task grid	Samples	Steps / LR	Seeds	Primary purpose
E1	dense linear / 96	$\rho \in \{1, 2, 4, 8\}$, lag $\in \{0, 2, 4, 8\}$	256	500 / .01	4–11	causal phase grid
E2	dense linear / 32–256	$(1, 0), (2, 2), (4, 4), (8, 8)$	1024	400 / .01	0–7	empirical reference
E-NL	tanh, GRU / 32	$\rho = 4$, lag 2	256	1000 / .01	20–27	direct-drift certificate
Tangent pilot	tanh, GRU / 32	$\rho = 4$, lag 2	256	1000 / .01	1×8	five-output falsification
Tangent factorial	tanh, GRU / 32	$\rho = 4$, lag 2	256	1000 / .01	4×8	crossing-event audit
E3	dense linear / 96	$\rho = 4$, lag 2	256	500 / .01	40–47	CDC ablation

Commands for new runs under the current source. The following commands execute new runs from the repository root; they do not bitwise reproduce historical dirty-source outcomes:

```
python run_experiment.py e1 --config configs/e1_dense_rerun.yaml
python run_experiment.py e2 --config configs/e2_corrected_validation.yaml
python run_experiment.py e2r --config configs/e2r_solver_checks.yaml
python run_experiment.py enl --config configs/enl_tanh_crossover.yaml
python scripts/build_enl_learnability_profile.py
python run_experiment.py e3 --config configs/e3_cdc_dense_ablation.yaml
```

The checked-in frozen evaluate configs are historical evidence records: they pin the original preflights and intentionally fail against later source. A new protocol run must first execute `enl-preflight`, then copy the matching evaluate config and explicitly repin `evaluation.preflight_dir` and `evaluation.manifest_sha256` before `enl-evaluate`. This creates a new study; it does not bitwise reproduce the archived dirty-source studies. The broad pilot pins manifest SHA-256 `f186d587...f0ffaf5` and prediction SHA-256 `9787d999...f9dfaa`; its evaluation verdict is failed. The restricted factorial pins manifest SHA-256 `bdcf02ab...a0db1` and prediction SHA-256 `c05b3181...6542`; its crossing-only verdict is passed. The exploratory E-NL command is deliberately omitted from the confirmatory list; its complete configuration and negative result remain archived in the repository. The learnability-profile command is a read-only derivation from the tracked E-NL summary and does not run training.

D Frozen-tangent falsification and restricted follow-up

The sealed E19 pilot tested one preregistered conjunction over five outputs and 16 architecture–seed records. Correct classifications were phase 12/16, drift crossing 14/16, response crossing 14/16, causal certificate 14/16, and weak-only learnability 12/16. The conjunction failed because tanh learnability was 4/8, below the frozen 5/8 floor; GRU learnability was 8/8. Magnitude calibration also differed sharply by architecture: tanh response-gap, weak-response, and drift RMSE were 2.367, 2.394, and 0.486, versus 0.0163, 0.0281, and 0.0107 for GRU. These numbers are preserved as a falsification result rather than averaged into a pass.

E20 was specified only after E19’s rejection and used fresh crossed seeds. Its accepted scope was restricted to event labels: response crossing was correct in 64/64 records and drift crossing in 60/64 (32/32 tanh, 28/32 GRU), with two-way bootstrap intervals $[1, 1]$ and $[0.8125, 1]$. The rejected secondary outputs were

Table 6: Authoritative artifact map. Paths are relative to the repository root.

Claim family	Artifact directory
Controlled dense phase grid	<code>results/e1_dense_rerun-20260823-085142</code>
Finite-width empirical closure	<code>results/e2_corrected_validation-20260821-224630</code>
Exact special solver checks	<code>results/e2r_solver_checks-20260823-084911</code>
Final nonlinear crossover and β audit	<code>paper/artifacts/enl_tanh_crossover-20260824-141207</code>
Broad frozen-NTK pilot (failed)	<code>paper/artifacts/enl_ntk_pilot-20260825</code>
Restricted frozen-NTK factorial (crossing labels passed)	<code>paper/artifacts/enl_ntk_crossing_factorial-20260825</code>
Exploratory nonlinear audit	<code>results/enl_exact_regime_search-20260824-105003</code>
Dense CDC ablation	<code>results/e3_cdc_dense_ablation-20260823-090702</code>

substantially weaker: phase 52/64, causal certificate 56/64, and learnability 49/64, including tanh learnability 17/32. Predictions were systematically early (overall drift/response bias $-0.822/-1.189$), while tanh response-gap, weak-response, and drift RMSE were 2.455, 2.489, and 0.490. Consequently E20 is an uncalibrated crossing-event classification result at one controlled cell, not a phase, causal, learnability, time, magnitude, prevalence, or kernel-stability validation. Both compact archives preserve these scores and seals but not the unavailable exact dirty source snapshots, so integrity auditability does not imply historical replayability.

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